Nonparametric Curve Estimation HW1

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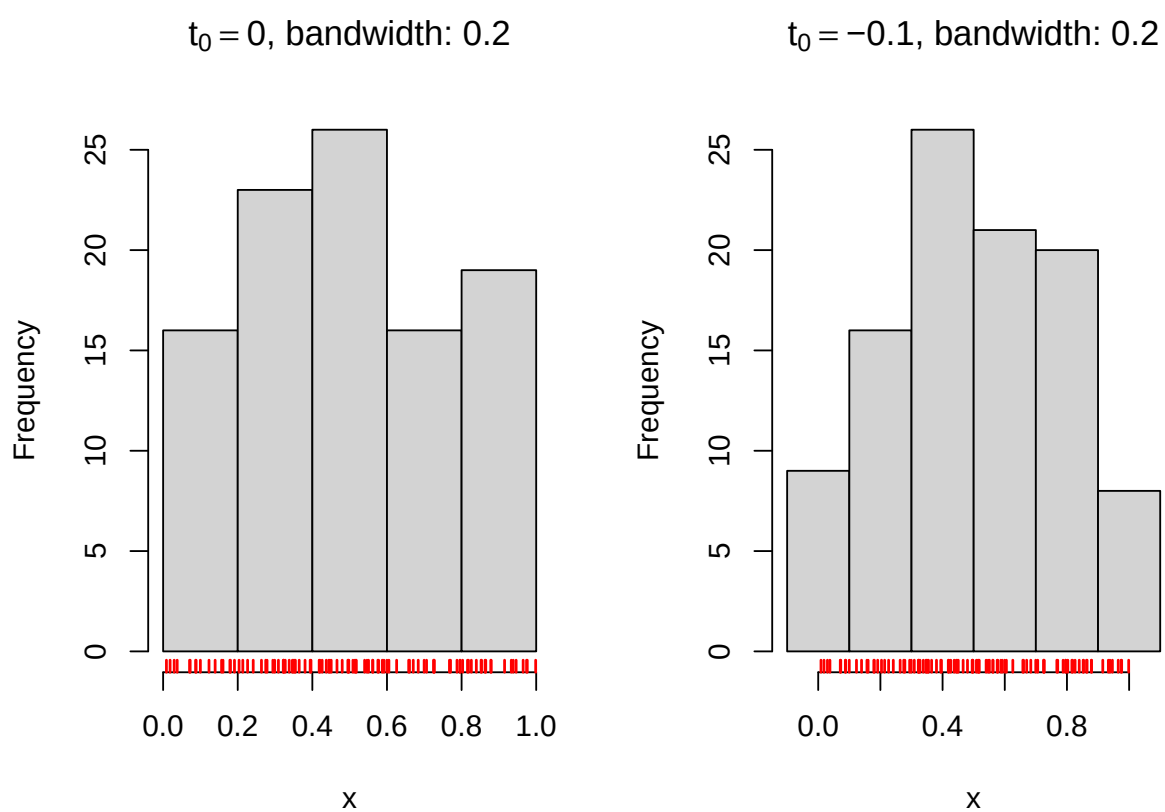
1.

(a)

```
set.seed(1234567)
x = runif(100)
t_0 = c(0, -0.1)
bin_width = c(0.2, 0.2)
par(mfrow = c(1, 2))
for(i in 1:2){
  hist(x, breaks = seq(t_0[i], max(x) + bin_width[i], by = bin_width[i]), border = "black",
       main = bquote(t[0] == .(t_0[i]) * ", bandwidth: " * .(bin_width[i])))
  rug(x, col = "red", lwd = 1.5)
}
```



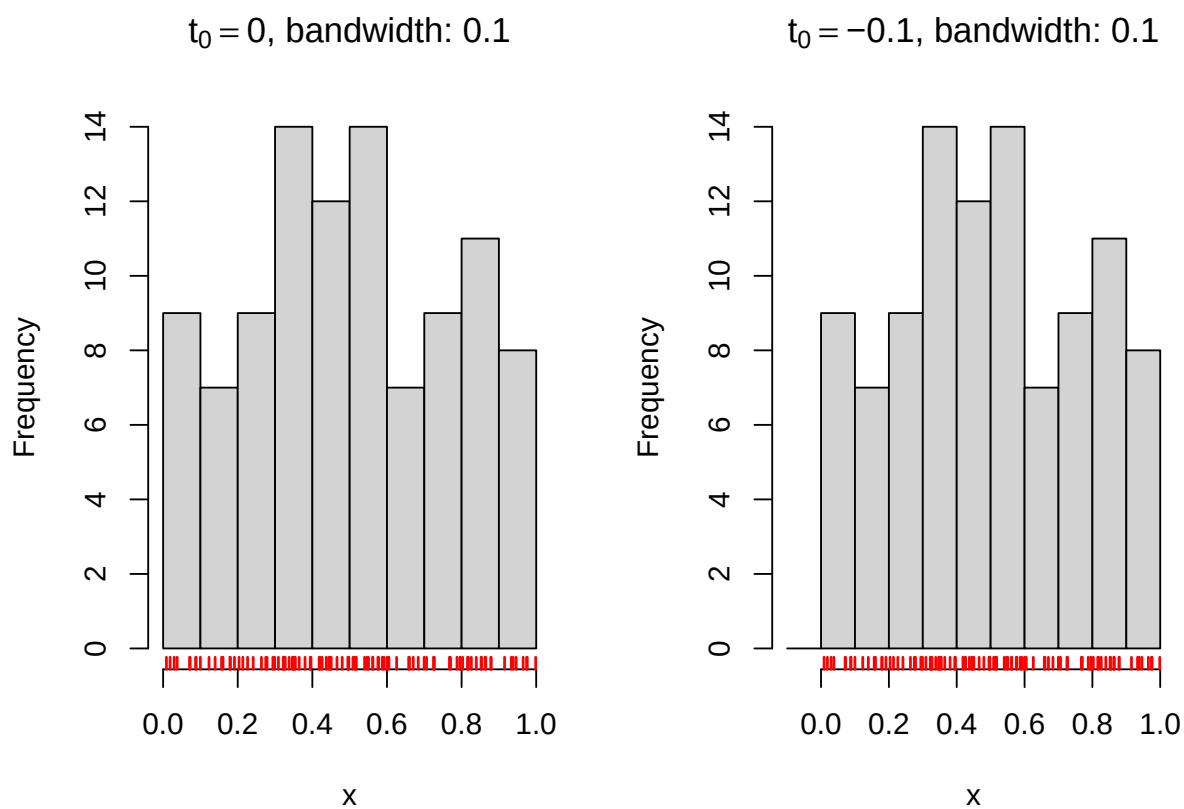
The first histogram with $t_0 = 0$ appears more uniform compared to the second histogram with $t_0 = -0.1$.

This difference arises because there are no observed sample points between -0.1 and 0. As a result, the first bin in the second histogram has a noticeably lower frequency than the other bins, making the distribution appear less uniform.

(b)

I change the bandwidth to 0.1.

```
bin_width = c(0.1, 0.1)
par(mfrow = c(1, 2))
for(i in 1:2){
  hist(x, breaks = seq(t_0[i], max(x) + bin_width[i], by = bin_width[i]), border = "black",
       main = bquote(t[0] == .(t_0[i]) * ", bandwidth: " * .(bin_width[i])))
  rug(x, col = "red", lwd = 1.5)
}
```



The first histogram with $t_0 = 0$ appears more uniform compared to the second histogram with $t_0 = -0.1$, for the same reason as in (a). Additionally, it is interesting to note that the histograms in (b) are both less smooth than those in (a), highlighting that the smoothness of a histogram depends on the bandwidth size.

2.

For (a) and (b), we first write down the log likelihood function assuming that the data follows a Beta distribution:

$$\ell(\alpha, \beta) = \sum_{i=1}^{n} \log f(x_i; \alpha, \beta)$$

where

$$f(x;\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

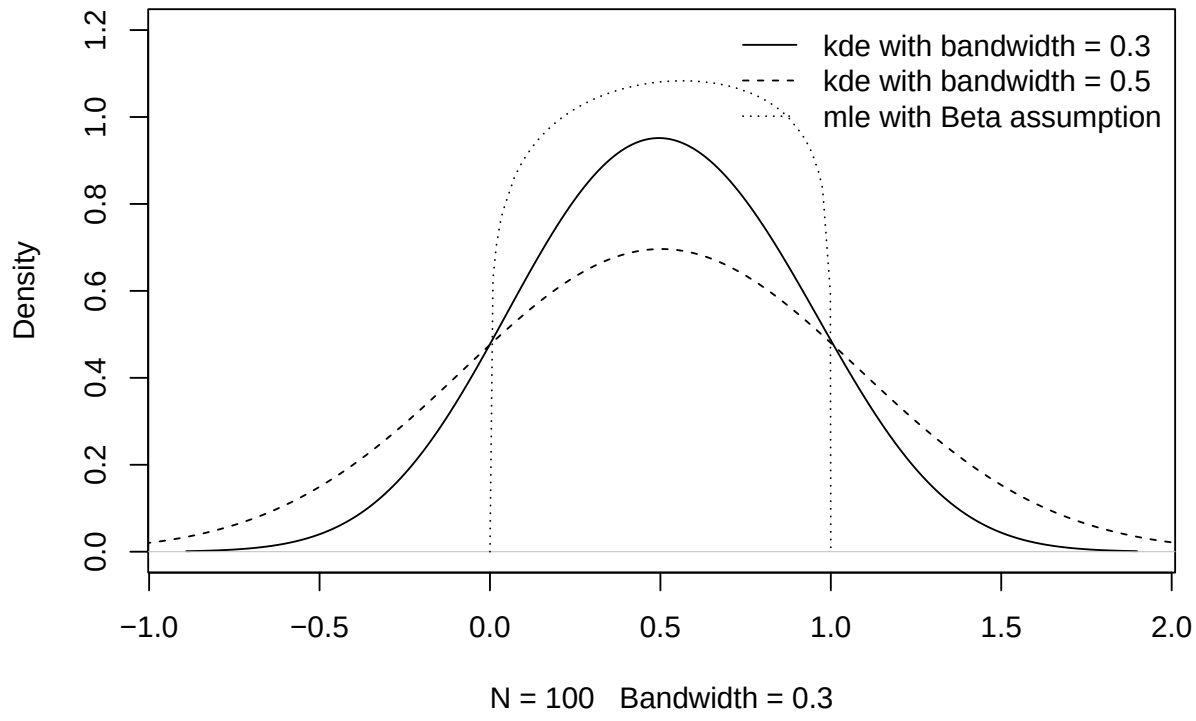
the density for the Beta distribution.

To solve the MLE of α and β, I implement the following code which obtain $(\hat{\alpha},\hat{\beta}) = \arg\max_{\alpha>0,\beta>0} \ell(\alpha,\beta)$.

```
minus_loglike  = function(params, x){
  -sum(dbeta(x, params[1], params[2], log = T))
}

opt = optim(par = c(0.1, 0.1),
      fn = minus_loglike,
      method = "L-BFGS-B",
      lower = 0,
      x = x)

alpha = opt$par[1]
beta = opt$par[2]

beta_density = dbeta(x, alpha, beta)
```

Subsequently, we can generate the plots required in (a) and (b) using bandwidths of 0.3 and 0.5.

```
bin_width = c(0.3, 0.5)
plot(density(x, kernel = "gaussian", bw = bin_width[1]),
     main = "", ylim = c(0, 1.2))
lines(density(x, kernel = "gaussian", bw = bin_width[2]), lty = 2)
lines(c(0, sort(x), 1), c(0, beta_density[order(x)], 0), lty = 3)
legend("topright", c("kde with bandwidth = 0.3",
                     "kde with bandwidth = 0.5",
                     "mle with Beta assumption"),
       lty = 1:3, bty = "n")
```



As observed, the kernel density estimate with a bandwidth of 0.5 is smoother than the one with a bandwidth of 0.3. However, neither kernel density estimate closely matches the estimates obtained under the Beta distribution assumption. Given that the true distribution of the data is uniform $(0, 1)$, the density estimate based on the Beta distribution assumption provides a better approximation of the true underlying density in this case. This because uniform $(0, 1)$ is in fact a special case of Beta distribution.

1 Source Code

```
knitr::opts_chunk$set(echo = TRUE, fig.width = 7, fig.height = 5)
library(kableExtra)
library(tidyverse)
library(knitr)
set.seed(1234567)
x = runif(100)
t_0 = c(0, -0.1)
bin_width = c(0.2, 0.2)
par(mfrow = c(1, 2))
for(i in 1:2){
  hist(x, breaks = seq(t_0[i], max(x) + bin_width[i], by = bin_width[i]), border = "black",
       main = bquote(t[0] == .(t_0[i]) * ", bandwidth: " * .(bin_width[i])))
  rug(x, col = "red", lwd = 1.5)
}
bin_width = c(0.1, 0.1)
par(mfrow = c(1, 2))
for(i in 1:2){
  hist(x, breaks = seq(t_0[i], max(x) + bin_width[i], by = bin_width[i]), border = "black",
       main = bquote(t[0] == .(t_0[i]) * ", bandwidth: " * .(bin_width[i])))
  rug(x, col = "red", lwd = 1.5)
}
minus_loglike  = function(params, x){
  -sum(dbeta(x, params[1], params[2], log = T))
}

opt = optim(par = c(0.1, 0.1),
      fn = minus_loglike,
      method = "L-BFGS-B",
      lower = 0,
      x = x)

alpha = opt$par[1]
beta = opt$par[2]

beta_density = dbeta(x, alpha, beta)
bin_width = c(0.3, 0.5)
plot(density(x, kernel = "gaussian", bw = bin_width[1]),
     main = "", ylim = c(0, 1.2))
lines(density(x, kernel = "gaussian", bw = bin_width[2]), lty = 2)
lines(c(0, sort(x), 1), c(0, beta_density[order(x)], 0), lty = 3)
legend("topright", c("kde with bandwidth = 0.3",
                     "kde with bandwidth = 0.5",
                     "mle with Beta assumption"),
       lty = 1:3, bty = "n")
```

Nonparametric Curve Estimation HW 0219

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1.

See above.

2.

See above.

3.

(a) Define $\theta_j = f(x)$ for $x \in B_j$

(b) Write the log likelihood:

$$\mathcal{L}(\theta_1, \dots, \theta_m) = \prod_{i=1}^{n} \prod_{j=1}^{m} \theta_j^{I(x_i \in B_j)}$$

$$\Rightarrow \ell(\theta_1, \dots, \theta_m) = \log \mathcal{L}(\theta_1, \dots, \theta_m)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} I(x_i \in B_j) \log \theta_j$$

(3) Impose the constraint:

$$\int \hat{f}(x)\, d\mu(x) = 1$$

$$\Leftrightarrow \sum_{j=1}^{m} \theta_j w_j = 1$$

(4) Maximize $\ell(\underset{\sim}{\theta})$ under the constraint:

By Lagrange multiplier, we want to maximize

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$$Q(\theta,\lambda)=\sum_{i=1}^{n}\sum_{j=1}^{m} I(x_i\in B_j)\log\theta_j+\lambda\left(\sum_{j=1}^{m}\theta_j w_j-1\right)$$

$$\frac{\partial}{\partial\theta_j}Q(\theta,\lambda)=\sum_{i=1}^{n} I(x_i\in B_j)\cdot\frac{1}{\theta_j}+\lambda w_j \overset{\text{set}}{=} 0 \iff \frac{f_j}{\theta_j}+\lambda w_j=0 \iff \theta_j=\frac{-f_j}{\lambda w_j} \quad \text{for } j=1,\dots,m$$

$$\frac{\partial}{\partial\lambda}Q(\theta,\lambda)=\sum_{j=1}^{m}\theta_j w_j-1 \overset{\text{set}}{=} 0 \quad \text{— ①}$$

Substitute $\theta_j=\frac{-f_j}{\lambda w_j}$ to ①,

$$\Rightarrow \sum_{j=1}^{m}\frac{-f_j}{\lambda \cancel{w_j}}\cancel{w_j}=1 \Rightarrow \lambda=-\sum_{j=1}^{m}f_j=-n$$

Substitute $\lambda=-n$ to $\frac{f_j}{\theta_j}+\lambda w_j=0$

$$\Rightarrow \frac{f_j}{\theta_j}-w_j n=0$$

$$\Rightarrow \theta_j=\frac{f_j}{n w_j}$$, which is the histogram estimator.

Therefore, the histogram estimator is the unique MLE over the set of f that are piecewise constant on B_j, $j=1,\dots,m$.

4. It suffices to show that $\int_{-\infty}^{\infty}\hat{f}(t)\,dt\neq 1$, where $\hat{f}(t)$ is a kNN estimator.

proof: $$\int_{-\infty}^{\infty}\hat{f}(t)\,dt=\int_{-\infty}^{\infty}\frac{k}{2n d_k(t)}\,dt=\lim_{b\to-\infty}\lim_{c\to\infty}\int_b^c\frac{k}{2n d_k(t)}\,dt$$

$$=\lim_{b\to-\infty}\lim_{c\to\infty}\int_b^{x_{(1)}}\frac{k}{2n(x_{(k)}-t)}\,dt+\int_{x_{(1)}}^{x_{(n)}}\frac{k}{2n d_k(t)}\,dt+\int_{x_{(n)}}^{c}\frac{k}{2n(t-x_{(n-k+1)})}\,dt$$

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$$= \lim_{b \to -\infty} \left[-\frac{k}{2n} \log\left(x_{(k)} - t\right) \Big|_{b}^{x_{(1)}} \right]$$

$$+ \int_{x_{(1)}}^{x_{(n)}} \frac{k}{2n\, d_k(t)}\, dt$$

$$+ \lim_{c \to \infty} \left[\frac{k}{2n} \log\left(1 - x_{(n-k+1)}\right) \Big|_{x_{(n)}}^{c} \right]$$

$$= \infty .$$

The proof is thus complete.

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